

The Abnormal Cardiovascular Examination and Clinical Decision Making

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Session Objectives

The student will:

- 1. Differentiate normal vs. abnormal findings of the cardiovascular history and physical examination.
- 2. Identify and distinguish abnormal examination findings including murmurs, rubs, and gallops.
- 3. Develop clinical decision-making process by analyzing and incorporating a patient's history and physical examination findings to formulate a thorough differential diagnosis, assessment, and plan.
- 4. Utilize appropriate SOAP note documentation methods relevant to the cardiovascular system.

Cardiovascular Examination

- **Vital signs**
 - HR, BP, BMI, Waist circumference
- **Neck**
 - JVP, Carotid arteries
- **Chest**
 - Cardiac
- **Abdomen**
 - Abdominal aorta
- **Lower extremities**
 - Pulses, edema
- **Special tests**

Vital Signs:

Heart Rate and Blood Pressure

- **HR:** normal = **60-100**
 - Tachycardia **>100**
 - Bradycardia **<60**
- **BP Classification**
→ **JNC 7 has evolved ... let's explore!**

JNC 7 Blood Pressure Classification in Adults Aged ≥ 18 Years			
Category	Systolic		Diastolic
Normal	<120	and	<80
Prehypertension	120-139	or	80-89
Hypertension, Stage 1	140-159	or	90-99
Hypertension, Stage 2	≥160	or	≥100

National Heart, Lung and Blood Institute. *JNC 7 Express. The Seventh Report of the Joint National Committee on the Prevention, Detection, Evaluation and Treatment of High Blood Pressure.* 2003

The Evolution of BP Classification Guidelines

2017 Guideline for the Prevention, Detection, Evaluation and Management of High Blood Pressure in Adults

BP Classification (JNC 7 and ACC/AHA Guidelines)

SBP		DBP	JNC 7	2017 ACC/AHA
<120	and	<80	Normal BP	Normal BP
120–129	and	<80	Prehypertension	Elevated BP
130–139	or	80–89	Prehypertension	Stage 1 hypertension
140–159	or	90–99	Stage 1 hypertension	Stage 2 hypertension
≥160	or	≥100	Stage 2 hypertension	Stage 2 hypertension

- Blood Pressure should be based on an average of ≥2 careful readings on ≥2 occasions
- Adults being treated with antihypertensive medication designated as having hypertension

Current BP Classification Guidelines

Blood Pressure Categories



BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)		DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120-129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130-139	or	80-89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
HYPERTENSIVE CRISIS (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

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heart.org/bplevels

Vital Signs:

BMI and Waist Circumference

CLASSIFICATION OF OVERWEIGHT AND OBESITY BY BMI, WAIST CIRCUMFERENCE AND ASSOCIATED DISEASE RISKS

	BMI (kg/m ²)	Obesity Class	Disease Risk* Relative to Normal Weight and Waist Circumference	
			Men ≤ 102 cm (≤ 40 in) Women ≤ 88 cm (≤ 35 in)	> 102 cm (> 40 in) > 88 cm (> 35 in)
Underweight	<18.5		—	—
Normal†	18.5 – 24.9		—	—
Overweight	25.0 – 29.9		Increased	High
Obesity	30.0 – 34.9	I	High	Very High
	35.0 – 39.9	II	Very High	Very High
Extreme Obesity	≥40	III	Extremely High	Extremely High

* Disease risk for type 2 diabetes, hypertension, and CVD.

+ Increased waist circumference can also be a marker for increased risk even in persons of normal weight.

Neck: Jugular Venous Pressure and Carotid Pulse

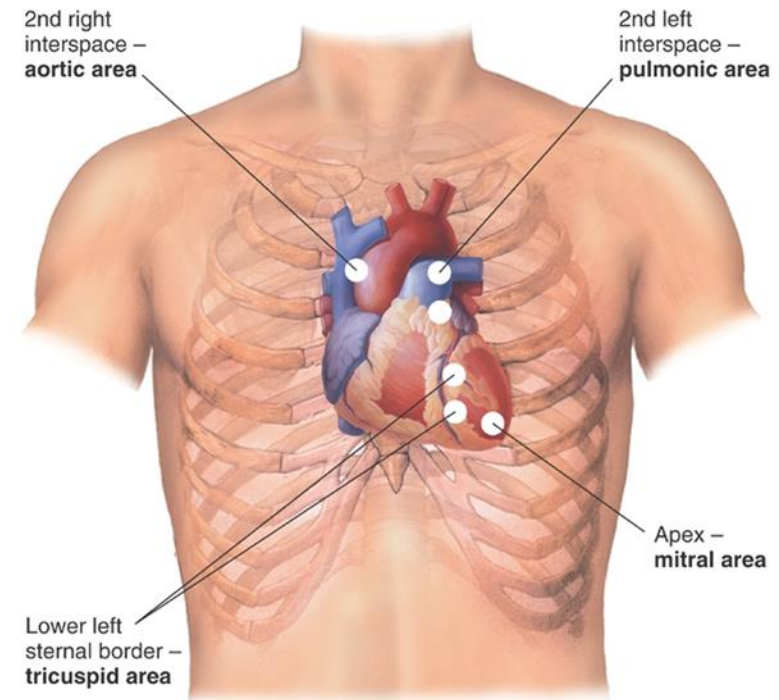
- JVP measured **>3cm (or possibly 4cm) above the sternal angle** → elevated
 - CHF, Pulmonary hypertension, tricuspid stenosis, pericardial compression/tamponade.
- JVP **falls** with hypovolemia and/or loss of blood
- Carotid **upstroke** may be:
 - **Brisk** – normal
 - **Delayed** – suggests aortic stenosis
 - **Bounding** – suggests aortic insufficiency
- Auscultate carotid to assess for **bruits**

The Cardiac Cycle of Systole and Diastole

- **Systole:** the ventricles **contract**
 - The right ventricle pumps blood into the pulmonary arteries (pulmonic valve is **open**)
 - The left ventricle pumps blood into the aorta (aortic valve is **open**)
- **Diastole:** the ventricles **relax**
 - Blood flows from the right atrium → right ventricle (tricuspid valve is **open**)
 - Blood flows from the left atrium → left ventricle (mitral valve is **open**)

Physical Exam - Auscultation

- Aortic Area
 - 2nd ICS, right sternal border
- Pulmonic Area
 - 2nd ICS, left sternal border
- Tricuspid Area
 - 3rd/4th ICS, left sternal border
- Mitral Area
 - 5th ICS, left midclavicular line



The Diaphragm and the Bell

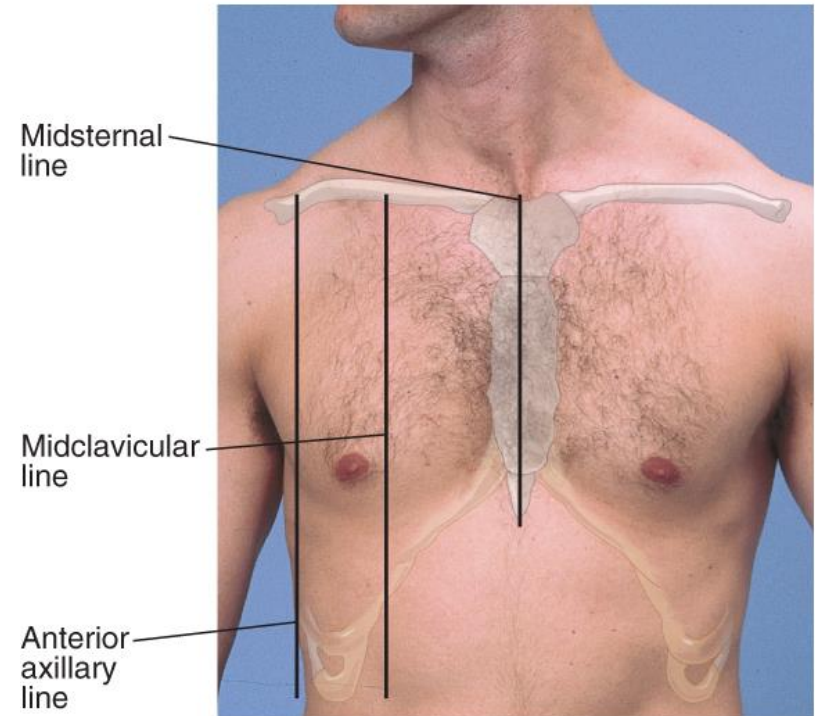
Copyright © 2007 Lippincott Williams & Wilkins. Instructor's Resource CD-ROM to Accompany Bates' Guide to Physical Examination and History Taking, ninth edition.

Cardiac Auscultation

- Auscultate in all listening areas for **S1** and **S2** using the **diaphragm** of the stethoscope
- Then auscultate at the apex with the bell
- The diaphragm and the bell ...
 - The **diaphragm** is best for detecting **high-pitched sounds** like S1, S2, and also S4 and most murmurs
 - The **bell** is best for detecting **low-pitched sounds** like S3 and the rumble of mitral stenosis

Know Your Surface Landmarks

- Count interspaces
 - Identify your ...
 - o Midsternal line
 - o Midclavicular line
 - o Anterior axillary line
 - o Midaxillary line



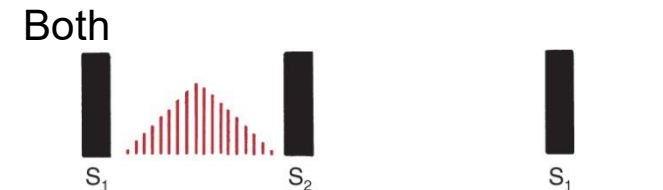
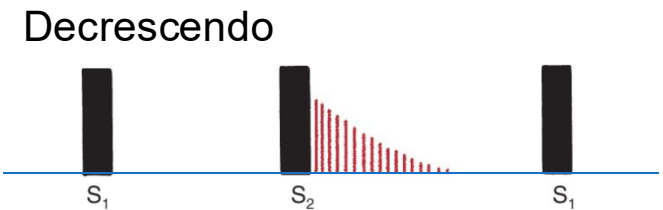
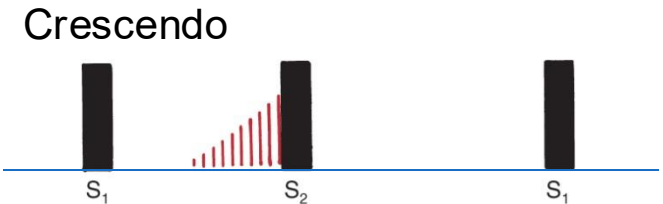


Describing Heart Murmurs: Timing and Duration

- Identify and describe any **murmurs**
- Timing: are the murmurs **systolic** or **diastolic**?
 - **Tip**: palpate the carotid upstroke (occurs in systole) as you auscultate
 - If the murmur coincides with the carotid upstroke, it is **systolic**
- Duration:
 - Early/mid/late systolic (or holosystolic)
 - Early/mid/or late diastolic

Describing Heart Murmurs: Shape

- Shape: Crescendo, decrescendo, or both (“diamond-shaped”)
 - Example: crescendo-decrescendo systolic murmur → **aortic stenosis**
 - Example: Holosystolic murmur → **mitral regurgitation**





Describing Heart Murmurs: Quality, Pitch, and Location

- Quality: Harsh, musical, soft, blowing, or rumbling?
- Pitch: High-, medium-, or low-pitched?
- Examples:
 - Harsh 2/6 medium-pitched holosystolic murmur best heard at the apex → **mitral regurgitation**
 - Soft, blowing 3/6 decrescendo diastolic murmur best heard at the lower left sternal border → **aortic regurgitation**

Describing Heart Murmurs: Intensity

Gradations of Murmurs

Grade	Description
Grade 1	Very faint, heard only after listener has “tuned in”; may not be heard in all positions
Grade 2	Quiet, but heard immediately after placing the stethoscope on the chest
Grade 3	Moderately loud
Grade 4	Loud, with <i>palpable thrill</i>
Grade 5	Very loud, with <i>thrill</i> . May be heard when the stethoscope is partly off the chest
Grade 6	Very loud, with <i>thrill</i> . May be heard with stethoscope entirely off the chest

Approach to Diagnosing Heart Murmurs: Timing and Duration

Approach to heart murmurs by description

Systolic or diastolic murmur?

(Differentiated by auscultation)

Systolic

Diastolic

Crescendo-decrescendo



Stenosis

Aortic or
pulmonary
stenosis

Holosystolic



Regurgitation

Mitral or
tricuspid
regurgitation

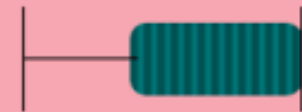
Early diastolic
or decrescendo



Regurgitation

Aortic or
pulmonary
regurgitation

Mid-late diastolic



Stenosis

Mitral or
tricuspid stenosis

* Continuous: systolic and diastolic murmur = Patent ductus arteriosus

Assessing the Point of Maximal Impulse

- The PMI may be:
 - Tapping — **normal**
 - Sustained → **LVH [hypertension or aortic stenosis]**
 - Diffuse → **dilated ventricle [CHF or cardiomyopathy]**
- PMI location, amplitude, duration, and diameter:
 - <2.5cm → **normal**
 - Pregnancy → **may displace PMI up and to the left**
 - RVH → **may displace PMI to the right**
 - LVH/Cardiomegaly → **may displace PMI laterally**
 - Pressure overload at LV [aortic stenosis] or volume overload at LV [mitral regurgitation] → **increased amplitude**

Abdomen: Abdominal aorta



- $<3\text{cm}$ → **normal**
- $>3\text{cm}$ → **abdominal aortic aneurysm**

Lower Extremities: Edema



- 0+ No pitting edema
- 1+ Mild pitting edema. 2 mm depression that disappears rapidly.
- 2+ Moderate pitting edema. 4 mm depression that disappears in 10–15 seconds.
- 3+ Moderately severe pitting edema. 6 mm depression that may last more than 1 minute.
- 4+ Severe pitting edema. 8 mm depression that can last more than 2 minutes.

Peripheral Arterial Disease and Ankle-Brachial Index

- ABI → specific and sensitive metric for the diagnosis of PAD.
- ABI has also been shown to predict mortality and adverse cardiovascular events independent of traditional CV risk factors.
- The major cardiovascular societies advise measuring an ABI in every smoker over 50 years old, every diabetic over 50, and all patients over 70.

$$\text{ABI} = \frac{\text{systolic pressure at the ankle}}{\text{systolic pressure at the arm}}$$

ABI Value	Interpretation	Recommendation
Greater than 1.4	Calcification / Vessel Hardening	Refer to vascular specialist
1.0 - 1.4	Normal	None
0.9 - 1.0	Acceptable	
0.8 - 0.9	Some Arterial Disease	Treat risk factors
0.5 - 0.8	Moderate Arterial Disease	Refer to vascular specialist
Less than 0.5	Severe Arterial Disease	Refer to vascular specialist

Clinical Problem Solving: Cardiovascular Complaints



- Always consider life threatening diagnosis
 - Angina
 - Myocardial infarction
 - Dissecting aortic aneurysm
 - Pulmonary embolism
- Etiology *may not be cardiac*
 - GI
 - Pulmonary
 - Musculoskeletal

Cardiac Etiology → Chest Pain

- Left sided chest pain
 - Chest pressure/tightness, sharp pain
- Shortness of breath
- Radiation to the jaw
- Nausea/Vomiting
- Diaphoresis



Atypical Presentation and Symptoms

- Women, geriatric and diabetic patients:
 - Upper back pain
 - Jaw pain
 - Arm pain
 - Shortness of breath
 - Paroxysmal nocturnal dyspnea
 - Nausea/vomiting
 - Fatigue

Understanding the symptoms – Palpitations

- Unpleasant awareness of the heartbeat
- Patients may describe palpitations
 - Skipping
 - Racing
 - Fluttering
 - Pounding
 - Stopping of the heart

- **Cardiac** etiologies
 - Irregular heart beat/arrhythmia
 - Forceful cardiac contraction
- Common *non-cardiac* etiologies
 - Anxiety
 - Hyperthyroidism

- **Dyspnea**: Uncomfortable awareness of breathing that is inappropriate to a given level of exertion
- **Orthopnea**: dyspnea that occurs when the patient is lying down and improves when patient sits up
- **Paroxysmal nocturnal dyspnea**: episodes of sudden dyspnea and orthopnea that awakens the patient from sleep

- Accumulation of excessive fluid in extravascular interstitial space
- Interstitial tissue can absorb liters of fluid
 - “Have you noticed any swelling?”
 - “Where does it occur?”
 - “Is it worse in the morning or at night?”

Assessing Cardiovascular Risks From Patient History

- Past Medical History
 - Dyslipidemia, hypertension, diabetes
- Past Surgical History
 - Previous cardiac surgeries
- Medications/Allergies
 - Compliance with medications
- Family History
 - FH of CVD and age of diagnosis
- Social History
 - Smoking - previous or active
 - Other drug use - previous or active
 - Physical inactivity
 - Unhealthy diet
- Vitals: Weight/BMI/Waist circumference
 - Obesity- especially central

Clinical Case and Chief Complaint

- Setting: Family Medicine Office
- **CHEST PAIN**



History of Present Illness

S.S. is a 76-year-old African-American male who presents to your office with a chief complaint of chest pain.

- How can you determine if you should continue with thorough history taking or call 911?



Clinical Decision-Making Pearl # 1

- What is the general appearance of the patient?
 - Is he in distress?
- What are the initial vital signs of the patient?
 - Is he hemodynamically stable?



Patient Appearance and ABCs

- Patient is sitting on exam table appearing worried and concerned, in no acute distress.
- T: 99.2°F
- HR: 96 bpm
- RR: 18
- B/P: 172/98
- SpO₂: 96% on RA



History of Present Illness

- What additional questions do you wish to ask for a complete HPI?
- What associated symptoms are you concerned about?



History of Present Illness

- **O**nset: 1 hour ago
- **P**alliative/**P**rovocative: Rest improves pain, activity worsens pain
- **Q**uality: Squeezing and pressure-like
- **R**adiation: Denies
- **S**everity: 9/10
- **T**iming: Constant; 1st episode
- What associated symptoms are you concerned about?



Chest Pain Associated Symptoms

- Chest trauma? Denies
- Recent travel? Denies
- Fevers or chills? Denies
- Diaphoresis? **Yes**
- Dyspnea? **Yes**
 - On exertion? **Yes**
 - Orthopnea? No
- Palpitations? No
- Leg swelling? No
- **Nausea/Vomiting? Yes**



Past Medical History and Past Surgical History

- Hypertension
- Dyslipidemia
- Asthma
- GERD
- Morbid obesity



Medications and Allergies

- Lisinopril
- Hydrochlorothiazide
- Amlodipine
- Simvastatin
- Albuterol inhaler
- Omeprazole
- Aspirin
- **NKDA**



Clinical Decision-Making Pearl # 2

- How can you ensure the patient is ***compliant*** with medications?
- How can this ***affect the management*** of patient's comorbidities?



Social History

- What questions would you like to ask about the patient's lifestyle and habits?



Social History

- **Tobacco use?** Former 20 pack year tobacco history; quit last week
- **Alcohol use?** 2 beers on weekend
- **Recreational drug use?** Denies
- **Caffeine intake?** 3-4 cups daily



Social History

- **Exercise?** Denies
- **Diet?** High in saturated fats
 - Eats out a lot, especially fast food
- **Occupation?** ER Nurse



Family History

- Patient reports he is adopted and does not know any familial medical history.



Review of Systems

- What systems would you concentrate on for a targeted ROS for this specific chief complaint?
- What symptoms would you ask about under each system?



Clinical Decision-Making Pearl # 3

- This is your 1st opportunity to target your history questions based on the differential diagnosis you are considering.
- Pertinent negative and positive review of systems will **help to “rule-in or rule-out”** diagnoses in your differential.



Review of Systems

- Constitutional: denies fevers, chills, night sweats, weight loss
- Cardiovascular: (+) *chest pain, diaphoresis, and dyspnea on exertion*; denies palpitations
- Respiratory: (+) *dyspnea on exertion*; denies cough
- Gastrointestinal: (+) *nausea*; denies vomiting, heartburn, abdominal pain
- Musculoskeletal: denies back pain, reproducible chest pain, joint pains
- Psychiatric: (+) *high stress level*; denies anxiety



Patient's Cardiac Risk Factor(s)

- What are the patient's specific cardiac risk factor(s)?
 - Modifiable?
 - Non-modifiable?



Clinical Decision-Making Pearl # 4 → Patient Education Opportunity

- Modifiable:

- Former smoker
- Hypertension
- Dyslipidemia
- Obesity
- Poor diet
- Sedentary lifestyle
- Stress

- Non-modifiable:

- Age
- Gender
- Family History



Physical Examination

- What systems would you like to target during physical examination?



Physical Examination

- Vital Signs: as above
- Ht: 70" Wt: 280 lbs BMI: 40.2 WC: 48"
- General: NAD
- Neck: Supple, no LAD, no JVD, no bruits
- Heart: Normal S1 S2, RRR, no M/R/G
- Lungs: CTA B/L, no W/R/R
- Abdomen: Soft, obese, NT/ND, normal BS, no pulsatile mass
- Extremities: no C/C/E



Laboratory and Diagnostic Imaging

- What in-house testing can aid in your diagnosis?



Laboratory and Diagnostic Imaging

- ECG
- CXR



Pertinent Components of History Taking In Patient with Chest Pain

- Important to quantify the patient's baseline level of activity!
 - Does pain occur with climbing stairs?
 - How many flights?
 - Does pain occur with walking?
 - 50 feet? 1 block?
 - Does pain occur with activities of daily living (minimal exertion)?
 - Carrying groceries? Vacuuming?



Pertinent Components of History Taking In Patient with Chest Pain

- How does this compare with these activities in the past?
- Are patient's **avoiding certain activities** due to pain or associated symptoms?
- **Detailed social history** (diet, caffeine, etc.)
 - **Ask patient to describe their diet!**
 - What do you typically eat for breakfast, lunch, dinner?
 - Do you feel “uncomfortably stuffed” after eating? (Portion-size)
 - What do you typically snack on?



What are your next steps?

- What is your differential diagnosis for chest pain?
- What is the most likely diagnosis?
- *How did you rule-in or rule-out other diagnoses in your differential?*
 - **In-house testing?**
 - Interpretation of ECG & CXR
 - **History and Physical exam findings?**



Assessment and Plan

- What initial labs did you order and why?
 - What follow-up labs did you order and why?
- How would you treat this patient's comorbid conditions?
 - Goals of treatment?
- What areas would you find somatic dysfunctions?
 - Goals of treatment?



Assessment/Plan: Preventative Medicine

- Consider counseling on the following topics:
 - Smoking cessation
 - Dietary modification(s)
 - Physical activity: goal → minimum 150 mins/wk.
- Consider comorbidities:
 - Diabetes, hypertension, and/or dyslipidemia
- Set a follow-up appointment:
 - Reassess risk factors, continue counseling, OMT

Prevention

Counseling

- **Cardiac Viscerosomatic Reflexes:** T1-5 on left
 - Acute viscerosomatic reflexes:
 - Warm, moist, boggy, edematous, +skin drag &/or +erythema tests
 - Chronic viscerosomatic reflexes:
 - Firm, dry sponginess, joint is “really stuck”, rubbery end-point
- **Cardiac Chapman's Reflexes:**
 - Anterior: Left 2nd ICS (between ribs 2&3) near sternum
 - Posterior: between the TPs of T2 and T3

Cardiac Sympathetic/Parasympathetic Innervation

Sympathetic: T1-T5

- Synapsing with thoracic and cervical chain ganglia

Parasympathetic: Vagus nerve

- Right Vagus innervates the SA node
- Left Vagus innervates the AV node

Sample Documentation of Your Physical Exam

O:

- **Vital Signs:**

- **Blood Pressure** (sitting): 110/70 right arm; 108/68 left arm.
- **Radial pulse**: 60 beats per minute, regular.

- **Neck:** No jugular venous distension; No carotid bruits.

- **Heart:** Normal S1, S2; regular rate and rhythm, no murmurs, rubs, or gallops.

- **Extremities:** No cyanosis, clubbing, edema.

O: (*w/ acceptable abbreviations*)

- **Vital Signs:**

- **BP** (sitting): 110/70 (R) arm; 108/68 (L) arm.
- **Radial pulse**: 60 bpm, regular.

- **Neck:** No JVD; No carotid bruits.

- **Heart:** Normal S1, S2; RRR; no M/R/G.

- **Extremities:** No C/C/E.

Reminders for Lab Session

- **Arrive promptly** to your assigned lab session and know which room you will be in.
- Wear **proper attire** for DPR lab.
- Bring your equipment:
 - Lab coat and appropriate DPR Attire
 - Stethoscope & BP cuff
 - Watch with a second hand
 - Penlight



Lecture Feedback Form

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

Thank You 😊



Reference Slides



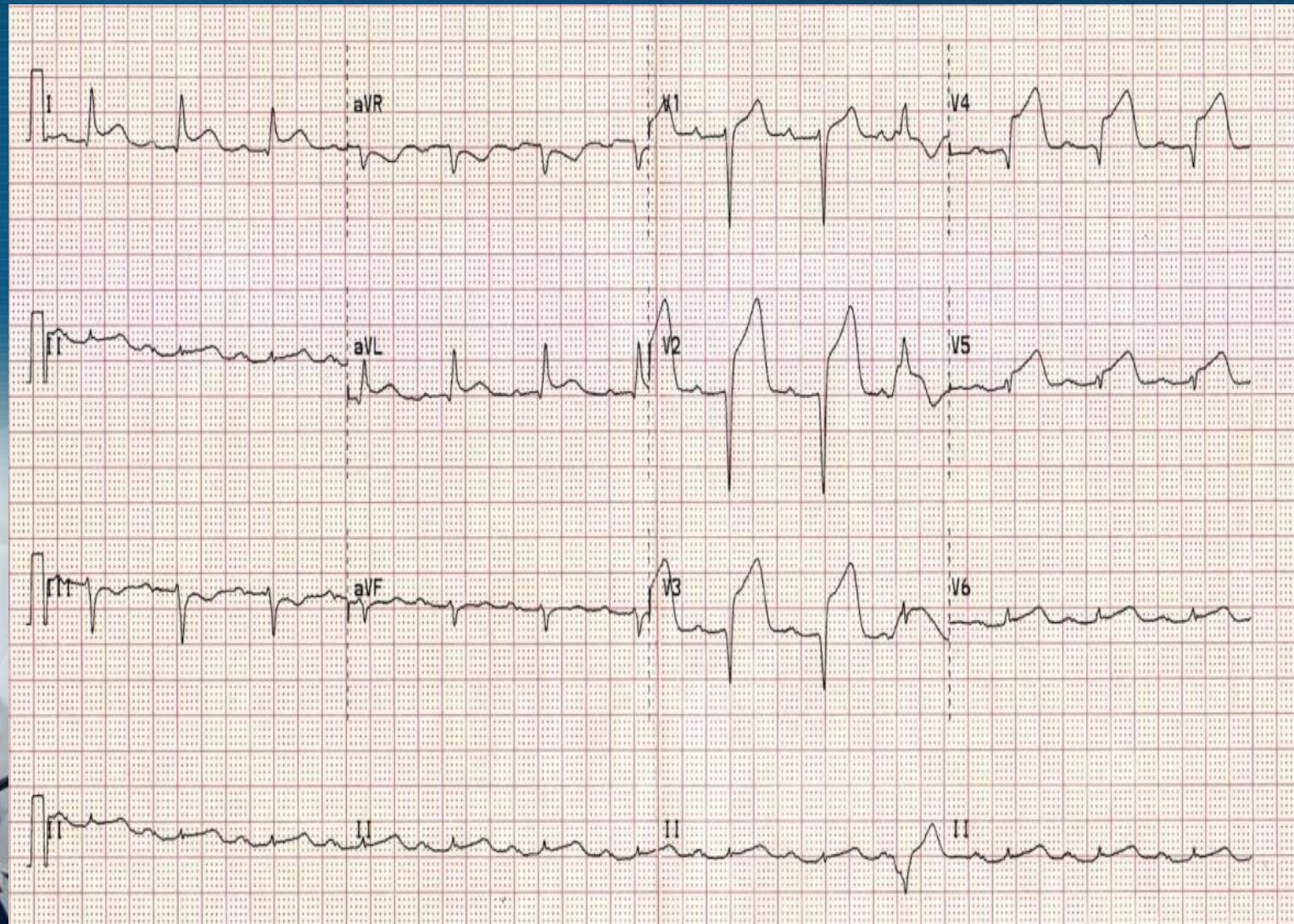
Heart Sound Libraries

<https://www.easyauscultation.com/>

<https://www.practicalclinicalskills.com/heart-sounds-murmurs>



ECG – Please Interpret

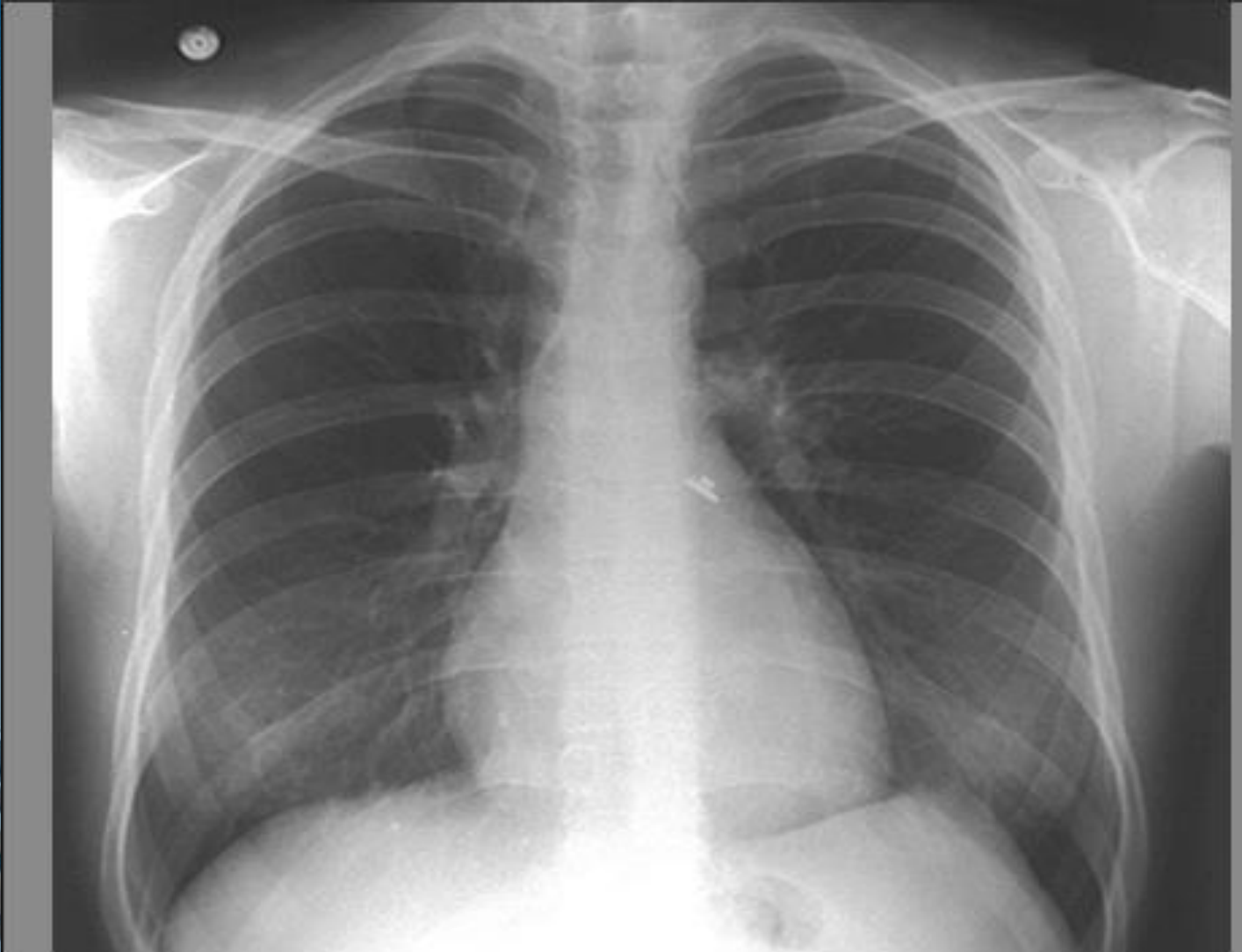


Initial Steps to Interpret an ECG

- Is this a sinus rhythm?
- What is the rate?
- What is the axis?
- Are there any ST or T-wave abnormalities?



CXR – Please Interpret



How to Read a CXR: ABCDE Method

- A: **A**irway
- B: **B**ones
- C: **C**ardiac (& Mediastinum) Silhouette
- D: **D**iaphragm
- E: **E**verything else (Lung fields & Lines)

